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PRESSURIZED SLEEVE FOR ROUTING OF LINEAR COMPONENTS

Abstract

According to an embodiment of the present invention, a pressurized sleeve for routing of linear components is provided. The pressurized sleeve is disposed to at least partially surround a flexible line object. A plurality of pouches is disposed in the robotic cable sleeve. At least two pouches of the plurality of pouches have different predetermined hydrostatic pressures from one another which induce axial movement of at least a portion of the robotic cable sleeve and a corresponding portion of the at least partially surrounded linear component.

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Background/Summary

BACKGROUND

[0001] Exemplary embodiments of the present invention relate to routing of linear components, and more particularly, to a pressurized sleeve for routing of linear components.

[0002] A project including installation and/or uninstallation (installation/uninstallation) of linear components (e.g., wires, cables, ropes, etc.) in relation to an overall apparatus can be a complex and tedious process involving a multitude of factors (e.g., project element characteristics/project environment characteristics/risks, etc.). The multitude of factors require precise analysis and modifications at various stages for success (e.g., before/during/after implementation). Failure to precisely anticipate/recognize/mitigate factors, such as excessive physical manipulation of the linear components, spatial conflicts between project elements, and/or inevitable deviations in project design frequently undermine project success and/or efficiency (e.g., installation/uninstallation of the linear components and/or other project elements).

[0003] For example, cabling is the most time-consuming and error-prone process in assembling a mainframe. Cabling is primarily performed by a human operator. Installation/uninstallation of cables to a mainframe without introducing mutual interference and/or damage proves to be a persistent difficulty for human operators alone. Even with proper instructions and training, cabling performed by human operators remains significantly flawed. Human operators are incapable of comprehensively negotiating the complex interplay of the multitude of factors, avoiding human error, recognizing imperceivable risks, performing ad hoc improvisations, avoiding damage to the cables/other project elements from excessive bending, and/or avoiding spatial conflicts due to selected arrangement and inevitable deviations therefrom.

[0004] Although apparatuses exist for installing certain project components (e.g., for routing individual wires in a substantially linear, single plane on a circuit board), these apparatuses can involve significant human input, elaborate mechanical machinery (which are prone to damage and/or are not feasible for a multitude of larger scale installations such as cabling), and neglect a precise analysis/implementation based on the complex interplay of the multitude of factors. Existing apparatuses attempt to address aspects of the stated problem but fail to facilitate completely autonomous installations/uninstallations for a variety of linear components, particularly based on a precise analysis of the multitude of installation/uninstallation factors and risks.

[0005] As a result, human operators and/or pre-existing apparatuses persist to perform installation/uninstallation of linear components with improper techniques, project designs, and/or applied forces, resulting in spatial conflicts, and/or actual/latent damage to project elements (e.g., the linear component, other project elements (e.g., component ports, cable connectors, other linear components, etc.), the apparatus for installation/uninstallation, etc.), and/or loss of overall system efficiency/functionality/lifespan.

SUMMARY

[0006] According to an embodiment of the present invention, a pressurized sleeve for routing of linear components is provided. A linear component is disposed at least partially surrounded by the pressurized sleeve. A plurality of pouches are disposed in the pressurized sleeve. At least two pouches of the plurality of pouches have different predetermined hydrostatic pressures from one another which induce movement in at least one dimension of at least a portion of the pressurized sleeve and a corresponding portion of the at least partially surrounded linear component.

[0007] According to an exemplary embodiment of the present invention, a pressurized sleeve for routing linear components is provided. A robotic cable sleeve is disposed at least partially surrounding a flexible linear component. A plurality of pouches and a plurality of sensors are disposed in the robotic cable sleeve, wherein at least two of the plurality of pouches have different predetermined hydrostatic pressures from one another that induce or preserve a predetermined shape of at least a portion of the robotic cable sleeve and a corresponding portion of the at least partially surrounded flexible linear component.

[0008] According to an exemplary embodiment of the present invention, a pressurized sleeve for routing linear components is provided. A robotic cable sleeve is disposed at least partially surrounding an at least partially flexible linear component. The at least partially flexible linear component is a wire or a cable, and the robotic cable sleeve includes a plurality of sensors and a plurality of pouches with adjustable hydrostatic pressures. A robotic stand is connected to the robotic cable sleeve. A controller is connected to the robotic cable sleeve and the robotic stand for implementing a predetermined project design. The implementing of the project design includes adjusting hydrostatic pressures of at least some of the pouches with adjustable hydrostatic pressures of the plurality of pouches with adjustable hydrostatic pressures. The implementing of the predetermined project design includes installation or uninstallation of the at least partially flexible linear component in relation to an apparatus by inducing predetermined movements, shapes, and positions of the robotic cable sleeve and a corresponding portion of the at least partially flexible linear component. At least one a 3D depth camera and an IoT feed is connected to the controller for obtaining real-time project design characteristics that include the implementation of the predetermined project design. The controller modifies the predetermined project design based on an analysis of the obtained real-time project design characteristics that include the implementation of the predetermined project design.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The following detailed description, given by way of example and not intended to limit the exemplary embodiments solely thereto, will best be appreciated in conjunction with the accompanying drawings, in which:

[0010] FIG. **1**A and FIG. **1**B illustrate a side cross-sectional view and a top cross-sectional view of a pressurized sleeve **100** for linear component **101** installation/uninstallation, respectively, in accordance with exemplary embodiments of the present inventive concept.

[0011] FIG. **2** illustrates a perspective x-ray view of a pressurized sleeve **100** for linear component **101** installation/uninstallation, in accordance with an exemplary embodiment of the present inventive concept.

[0012] FIG. **3** illustrates a perspective x-ray view of a pressurized sleeve **100** for linear component **101** installation/uninstallation, in accordance with an exemplary embodiment of the present invention.

[0013] FIG. **4** illustrates a schematic view of a system implementation that includes a pressurized sleeve **100** for linear component **101** installation/uninstallation, in accordance with an exemplary embodiment of the present invention.

[0014] FIG. **5** illustrates a schematic view of a system implementation for a pressurized sleeve **100** for linear component **101** installation/uninstallation, in accordance with an exemplary embodiment of the present invention.

[0015] FIG. **6** illustrates a schematic diagram of a computing environment **600** including a pressurized sleeve for linear component installation/uninstallation program **650**, in accordance with an exemplary embodiment of the present inventive concept.

[0016] FIG. **7** illustrates a block diagram of components **751-753** included in the pressurized sleeve for linear component installation/uninstallation program **650**, in accordance with an exemplary embodiment of the present inventive concept.

[0017] FIG. **8** illustrates a flowchart of a method of linear component installation/uninstallation using pressurized sleeve **800**, in accordance with an exemplary embodiment of the present inventive concept.

[0018] It is to be understood that the included drawings are not necessarily drawn to scale/proportion. The included drawings are merely schematic examples to assist in understanding of the present invention and are not intended to portray fixed parameters. In the drawings, like numbering may represent like elements.

DETAILED DESCRIPTION

[0019] Exemplary embodiments of the present invention are disclosed hereafter. However, it shall be understood that the scope of the present invention is dictated by the claims. The disclosed exemplary embodiments are merely illustrative of the claimed invention. The present invention may be embodied in many different forms and should not be construed as limited to only the exemplary embodiments set forth herein. Rather, these included exemplary embodiments are provided for completeness of disclosure and to facilitate an understanding to those skilled in the art. In the detailed description, discussion of well-known features and techniques may be omitted to avoid unnecessarily obscuring the presented exemplary embodiments.

[0020] References in the specification to “one embodiment,” “an embodiment,” “an exemplary embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but not every embodiment may necessarily include that feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to implement such feature, structure, or characteristic in connection with other embodiments whether explicitly described.

[0021] In the interest of not obscuring the presentation of the exemplary embodiments of the present invention, in the following detailed description, some processing steps or operations that are known in the art may have been combined for presentation and for illustration purposes, and in some instances, may have not been described in detail. Additionally, some processing steps or operations that are known in the art may not be described at all. The following detailed description is focused on the distinctive features or elements of the present invention according to various exemplary embodiments.

[0022] As aforementioned, a project including installation/uninstallation of linear components (e.g., wires, cables, ropes, etc.) in an overall apparatus is presently a complex, tedious, inefficient, and error prone process which is primarily performed by human operators. Apparatus based solutions directed to at least partial automation of linear component installation involve substantially linear planes of movement, require significant degrees of human input, elaborate mechanical devices (which are not feasible for a multitude of larger scale installations (e.g., cabling), and neglect analysis of the complex interplay of the multitude of installation factors. Human operators and/or preexisting apparatuses thus fail to effectively and consistently negotiate the multitude of factors, such as: connector styles, connection forces (e.g., plug forces), thresholds of predetermined use, project element characteristics (e.g., dimensions, materials, minimum/maximum bend radii, etc.); project environment characteristics (e.g., project designs, faithfulness, project workspaces, and inevitable deviations/improvisations, space limitations, destination points, etc.); risks (e.g., deviations in predetermined use thresholds, actual/latent project element damage, spatial conflicts between project elements, efficiency/functionality/lifespan of project elements/overall system, resource sufficiency, interference between installation/uninstallation devices and/or human operators, etc.), etc.

[0023] Embodiments of the present invention address these shortcomings and more with a pressurized sleeve for routing of linear components that includes pouches with adjustable hydrostatic pressures, which can be coupled with system implementations (e.g., controllers, 3D depth cameras, IoT feed, external computing devices, etc.) and methods therefor. Thus, the pressurized sleeve, system implementations, and methods therewith enable at least substantially autonomous linear component installation/uninstallation that avoids the identified shortcomings, at least.

[0024] FIG. **1**A and FIG. **1**B illustrate a side cross-sectional view and a top cross-sectional view of a pressurized sleeve **100** for linear component **101** installation/uninstallation, respectively, in accordance with exemplary embodiments of the present invention.

[0025] The pressurized sleeve **100** (also referred to herein as the robotic cable sleeve **100**) can be disposed to at least partially surround at least one linear component **101**. The linear component **101** can be loaded/fed into the pressurized sleeve **100**. The linear component **101** can be a wire, cable (e.g., electronic, fiber-optic, mechanical, etc.), rod, beam, rope, etc. The linear component **101** can be at least partially flexible. The pressurized sleeve **100** can include an elastic lining **107**. The elastic lining **107** can be at least partially flexible. The pressurized sleeve **100** can include at least one connection orientation fixture **108**. The pressurized sleeve **100** can include a plurality of pouches **102**. The plurality of pouches **102** can be disposed in and/or on the pressurized sleeve **100**. The plurality of pouches can be spaced in predetermined intervals and/or predetermined arrangements/configurations (groups/pairs/clusters/shapes) in at least one dimension/plane (e.g., a circular grouping configuration extending along a cross-sectional circumference of the pressurized sleeve **100** and/or rows extending in a lengthwise direction of the pressurized sleeve **100**). The plurality of pouches **102** can have volumes with hydrostatic pressures that are discrete from one another and/or are at least partially integrated (e.g., based on the predetermined arrangements/configurations). The pressurized sleeve **100** can include a plurality of sensors **103** (e.g., bend sensors, strain gauges, hydrostatic pressure sensors, vibrational sensors, audio sensors, etc.). The pressurized sleeve **100** can include and/or be connected to a pump **104** (e.g., pneumatic and/or hydraulic) and/or a compressor **105** (e.g., pneumatic and/or hydraulic). The pump **104** and the compressor **105** can be used cooperatively and/or alternatively as either an inlet or outlet. The connection orientation fixture **108** can be disposed on at least one terminal end of the pressurized sleeve **100**. The connection orientation fixture **108** can include a connection orientation fixture laser **109**. The connection orientation fixture laser **109** can facilitate linear component **101** installation/uninstallation (e.g., by detecting spatial movement/position/angle/orientation/connection site adjacency accuracy, etc.). The pressurized sleeve **100** and/or sub-components thereof (e.g., the sensors **103**, the air pouches **102**, the connection orientation fixture **108**, the connection orientation fixture laser **109**, the pump **104**, and/or the compressor **105**, etc.) can be connected to a controller **304** (shown in FIG. **3**) and/or an external computing device **405** (shown in FIG. **4**).

[0026] In the case of a pneumatic pump **104** and/or a pneumatic compressor **105**, various predetermined gases (e.g., air) can be used. In the case of a hydraulic pump **104** and/or a hydraulic compressor **105**, various predetermined fluids (e.g., water) can be used. The pouches **102** can be adjusted (e.g., inflated and/or deflated) according to predetermined hydrostatic pressures (e.g., calculated by the controller **304**). The hydrostatic pressures can be adjusted by the pump **104** and/or the compressor **105** (e.g., calculated/implemented by the controller **304** via electrical signal based on calculated predetermined hydrostatic pressures). The hydrostatic pressures of the pouches **102** can be adjusted uniformly, individually, and/or in arrangement/configurations (e.g., at least a partial circular configuration, opposing pairs, segments, etc.) to achieve a predetermined motion in at least one dimension (e.g., axial movement, elongation, contraction, curve, bend, twist, angle, orientation, position, shape, etc.) of the pressurized sleeve **100** and/or the corresponding portion of the at least partially surrounded linear component **101** in at least one dimension/plane. At least some of the pouches **102** can have predetermined hydrostatic pressures that are nonidentical to one another, transient, and/or static (e.g., permanent or premanufactured). Predetermined hydrostatic pressures can include opposing inflated/tension and deflated/compression regions **106***i*/**106***d*, respectively, which can be disposed substantially in parallel to one another.

[0027] The predetermined hydrostatic pressures can be a predetermined hydrostatic pressure of resistance that exceeds a predetermined (e.g., calculated by the controller **304** and/or determined based on the extracted features from the obtained data) capacity of physical manipulation/force

(e.g., bending) by human operators and/or installation/uninstallation apparatuses (e.g., the pressurized sleeve **100**). Excessive physical manipulation can refer to a force and/or induced motion with predetermined potential to cause damage to the linear component **101** (e.g., at/exceeds a minimum bend radius of the linear component **101**) and/or the pressurized sleeve **100**. At least some of the pouches of the plurality of pouches **102** can be calibrated to pop (e.g., audibly and/or by sensor **103** detection) when excessive physical manipulation occurs. The regions of tension/inflated and/or the compression/deflated **106***i*/**106***d* can, for example, create resistance to excessive bending of the corresponding portion of the at least partially surrounded linear component **101** and/or the pressurized sleeve **100**. The different predetermined hydrostatic pressures can cooperatively induce the predetermined motion of at least a portion of the pressurized sleeve **100** and the corresponding portion of the at least partially surrounded linear component **101**.
[0028] FIG. **2** illustrates a perspective x-ray view of a pressurized sleeve **100** for linear component **101** installation/uninstallation, in accordance with an exemplary embodiment of the present invention. The pressurized sleeve **100** can include at least one gas (e.g., air) and/or fluid (e.g., water) manifold **201**. The manifold **201** can be adjustably connected to the controller **304**. The manifold **201** can be connected to the compressor **105** and/or the pump **104**. The compressor **105** and/or the pump **104** can be further connected to a plurality of compressor lines **205** and/or a plurality of pump lines **204**, respectively. The compressor lines **205** and/or the pump lines **204** can be connected to at least some of the pouches **102** (e.g., discrete, at least partially integrated, and/or according to arrangement/configuration. The compression lines **205** and/or the pump lines **204** can resemble arterial connectivity to organic tissues. The at least one manifold **201** can include a plurality of manifold openings **206** and/or manifold apertures thereof. The apertures of the manifold openings **206** can be adjustably connected to the controller **304** (e.g., opened/closed, dilated/constricted, etc.), to achieve precise predetermined hydrostatic pressures of the predetermined pouches **102**.
[0029] FIG. **3** illustrates a perspective x-ray view of a pressurized sleeve **100** for linear component **101** installation/uninstallation, in accordance with an exemplary embodiment of the present invention. The pressurized sleeve **100** can include the sensors **103** (e.g., strain gauges and bend sensors). The sensors **103** can be arranged/configured as described/illustrated with respect to FIG. **1** and/or differently from one another as appropriate for predetermined measurement value obtainment. For example, the strain gauges **103***a* can be arranged differently from the bend sensors **103***b* (e.g., in a lengthwise configuration versus a circular/circumferential configuration). As aforementioned, the sensors **103** can be further connected to the controller **304**.
[0030] FIG. **4** illustrates a schematic view of a system implementation that includes the pressurized sleeve **100** for linear component **101** installation/uninstallation, in accordance with an exemplary embodiment of the present invention.
[0031] The system implementation of the pressurized sleeve **100** can be planned and/or occur in a predetermined project environment **400** such as a predetermined project space (e.g., a room). The project environment **400** can include a predetermined cabling workspace **404** of predetermined dimensions/characteristics, at least one 3D depth camera **406**, an external computing device **405** (can also be remotely located), and/or at least one IoT feed. The cabling workspace **404** can include the apparatus for installation/uninstallation of the linear component **101**. The robotic stand **401** and/or sub-components thereof (e.g., a base **402** and/or a fixture **403**) can be connected to the controller **304** and/or the external computing device **405**. The controller **304** and/or the external computing device **405** can be further connected to one another, the pressurized sleeve **100** and/or sub-components thereof (e.g., the at least one 3D depth camera **406** and/or the IoT feed (e.g., to get a 3D model of the project environment **400**, project environment dimensions, evaluate progress, to facilitate linear component **101** installation/uninstallation, detect deviations, etc. in real-time)), the at least one manifold **201**, the manifold openings **206**, and/or the manifold apertures thereof, etc.
[0032] The controller **304** and/or the external computing device **405** can perform feature extraction

on the obtained project data, analyze extracted features, and/or calculate logistics of project design implementation (e.g., installation/uninstallation), such as negotiating various installation factors (e.g., connector styles, linear component characteristics (e.g., flexibility, dimensions, types, materials, manufacturers, minimum/maximum bend radii, etc.), configuration characteristics (e.g., project designs and inevitable deviations therefrom, space limitations, destination points, etc.), plug forces, etc. The controller **304** can perform predetermined adjustments to the robotic stand **401** (e.g., to achieve a predetermined height, orientation, angle, and/or position, etc.), as calculated to be necessary for the predetermined project design implementation (e.g., installation/uninstallation of the linear component **101**), step(s), and/or dynamic modifications thereto. After the linear component **101** is loaded into the pressurized sleeve **100**, the controller **304** and/or the external computing device **405** can adjust predetermined hydrostatic pressures of the predetermined pouches **102** to achieve desired actions (e.g., movements (e.g., axial moving, routing, angling, bending, curving, orienting, positioning, etc.), spatial positions (e.g., of the linear component **101** and/or terminal ends (e.g., plugs) thereof (e.g., relative to connection sites)), 3D shapes, installation/uninstallation, etc.).

[0033] The robotic stand **401** can be adjustably moved (e.g., rotated/raised/lowered/moved/angled/oriented, etc.) mechanically, by the controller **304**, and/or the external computing device **405** (e.g., via electrical signal). For example, the base **402** can be equipped with a means for movement (e.g., pivots, bearings, wheels, treads, propellors, etc.). Adjustments to the robotic stand **401** can be calculated/implemented according to a predetermined project design and/or dynamic changes thereto. The fixture of the robotic stand **401** can include an adjustable sphincter (e.g., connected to the pump **104** and/or the compressor **105** for diametric expansion/contraction), clasp, ring, tie, etc. to secure/remove the linear component **101** and/or the pressurized sleeve **100**. The pressurized sleeve **100** can be loaded into the fixture **403**. The linear component **101** can be loaded into the pressurized sleeve **100**. The base **402** and/or the fixture **403** can be adjusted (e.g., tilted, asymmetrically elongated/contracted, etc.) to achieve a relative loaded position of the linear component **101** in relation to a predetermined point of the pressurized sleeve **100** and/or the robotic stand **401** (e.g., to ensure symmetrical loading relative to the fixture **403**).

[0034] In an embodiment, to get a greater quality 3D model of the project environment **400**, one or more of the sensors **103** can move along a track **103-1** (e.g., a linear guide track) to capture various perspectives (e.g., movements, angles, positions, spatial positions, orientations, 3D shapes, etc.).

[0035] The presence/absence of the robotic stand **401** and/or project elements in the predetermined project environment **400** can be determined by the controller **304** and/or the external computing device **405** from analyzed extracted features from real-time obtained project data (e.g., via the at least one 3D depth camera **406** and/or IoT feed). In an embodiment, when a project environment **400** does not include the robotic stand **401** and/or predetermined project elements upon initiation of a predetermined project, the controller **304** and/or the external computing device **405** can deploy the robotic stand **401** (e.g., with the pressurized sleeve **100** and/or the linear component **101**), orchestrate delivery of the predetermined project elements, modify the predetermined project environment **400**, and/or alert a predetermined user accordingly.

[0036] FIG. **5** illustrates a schematic view of a system implementation that includes a pressurized sleeve **100** for linear component **101** installation/uninstallation, in accordance with an exemplary embodiment of the present invention.

[0037] The linear component **101** can be loaded into the pressurized sleeve **100** (e.g., the robotic cable sleeve **100**) connected to the robotic stand **401** (e.g., by the fixture **403**). The initial inflation of at least some of the pouches **102** can be calculated to effectuate an axial movement of the pressurized sleeve **100** and/or at least partially surrounded linear component **101** such that they are substantially parallel to a plane of the ground for installation or vice versa in the case of uninstallation. The pressurized sleeve **100** can initially only partially surround the linear component **101** and can extend to terminal ends thereof upon the controller **304** and/or the external computing device **405** adjusting a predetermined inflation and/or deflation (e.g., predetermined hydrostatic pressures) of at least some predetermined pouches **102** via adjustments to the pump **104** and/or the compressor **105**. Upon net inflation of the at least some of the pouches **102**, the pressurized sleeve **100** can extend to at least one terminal end of the linear component **101** and can completely surround the body of the linear component **101** (or vice versa in the case of net deflation). In an embodiment, a predetermined buffer slack space can be calculated/provided at terminal ends of the linear component **101** such that further inflation of the pressurized sleeve **100** will provide a requisite predetermined connection force (e.g., the plug force) to complete installation. For example, further inflation of at least some pouches **102** in the terminal ends of the pressurized sleeve **100** can facilitate installation of the linear component **101** when the controller **304** and/or the external computing device **405** confirms a predetermined shape/spatial position/movement thereof based on the connection orientation fixture laser **109** of the connection orientation fixture **108**, the IoT feed, and/or the at least one 3D depth camera **406**. In an embodiment, the pressurized sleeve **100** can extend by a predetermined length beyond at least one terminal end of the linear component **101** to provide a predetermined hydrostatic pressure cushion for shock absorption (e.g., plugging, suspension, etc.).

[0038] FIG. **6** illustrates a schematic diagram of computing environment **600** including a pressurized sleeve for linear component installation/uninstallation program **650**, in accordance with an exemplary embodiment of the present inventive concept.

[0039] Various aspects of the present disclosure are described by narrative text, flowcharts, block diagrams of computer systems and/or block diagrams of the machine logic included in computer program product (CPP) embodiments. With respect to any flowcharts, depending upon the technology involved, the operations can be performed in a different order than what is shown in a given flowchart. For example, again depending upon the technology involved, two operations shown in successive flowchart blocks may be performed in reverse order, as a single integrated step, concurrently, or in a manner at least partially overlapping in time.

[0040] A computer program product embodiment ("CPP embodiment" or "CPP") is a term used in the present disclosure to describe any set of one, or more, storage media (also called "mediums") collectively included in a set of one, or more, storage devices that collectively include machine readable code corresponding to instructions and/or data for performing computer operations specified in a given CPP claim. A "storage device" is any tangible device that can retain and store instructions for use by a computer processor. Without limitation, the computer readable storage medium may be an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, a mechanical storage medium, or any suitable combination of the foregoing. Some known types of storage devices that include these mediums include: diskette, hard disk, random access memory (RAM), read-only memory (ROM), erasable programmable read-only memory (EPROM or Flash memory), static random access memory (SRAM), compact disc read-only memory (CD-ROM), digital versatile disk (DVD), memory stick, floppy disk, mechanically encoded device (such as punch cards or pits/lands formed in a major surface of a disc) or any suitable combination of the foregoing. A computer readable storage medium, as that term is used in the present disclosure, is not to be construed as storage in the form of transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide, light pulses passing through a fiber optic cable, electrical signals communicated through a wire, and/or other transmission media. As will be understood by those of skill in the art, data is typically moved at some occasional points in time during normal operations of a storage device, such as during access, de-fragmentation or garbage collection, but this does not render the storage device as transitory because the data is not transitory while it is stored.

[0041] Computing environment **600** contains an example of an environment for the execution of at least some of the computer code involved in performing the inventive methods, such as the pressurized sleeve for linear component installation/uninstallation program **650**. In addition to block **650**, computing environment **600** includes, for example, computer **601**, wide area network (WAN) **602**, end user device (EUD) **603**, remote server **604**, public cloud **605**, and private cloud **606**. In this embodiment, computer **601** includes processor set **610** (including processing circuitry **620** and cache **621**), communication fabric **611**, volatile memory **612**, persistent storage **613** (including operating system **622** and block **650**, as identified above), peripheral device set **614** (including user interface (UI) device set **623**, storage **624**, and Internet of Things (IoT) sensor set **625**), and network module **615**. Remote server **604** includes remote database **630**. Public cloud **605** includes gateway **640**, cloud orchestration module **641**, host physical machine set **642**, virtual machine set **643**, and container set **644**.

[0042] COMPUTER **601** may take the form of a desktop computer, laptop computer, tablet computer, smart phone, smart watch or other wearable computer, mainframe computer, quantum computer or any other form of computer or mobile device now known or to be developed in the future that is capable of running a program, accessing a network or querying a database, such as remote database **630**. As is well understood in the art of computer technology, and depending upon the technology, performance of a computer-implemented method may be distributed among multiple computers and/or between multiple locations. On the other hand, in this presentation of computing environment **600**, detailed discussion is focused on a single computer, specifically computer **601**, to keep the presentation as simple as possible. Computer **601** may be located in a cloud, even though it is not shown in a cloud in FIG. **6**. On the other hand, computer **601** is not required to be in a cloud except to any extent as may be affirmatively indicated.

[0043] PROCESSOR SET **610** includes one, or more, computer processors of any type now known or to be developed in the future. Processing circuitry **620** may be distributed over multiple packages, for example, multiple, coordinated integrated circuit chips. Processing circuitry **620** may implement multiple processor threads and/or multiple processor cores. Cache **621** is memory that is located in the processor chip package(s) and is typically used for data or code that should be available for rapid access by the threads or cores running on processor set **610**. Cache memories are typically organized into multiple levels depending upon relative proximity to the processing circuitry. Alternatively, some, or all, of the cache for the processor set may be located “off chip.” In some computing environments, processor set **610** may be designed for working with qubits and performing quantum computing.

[0044] Computer readable program instructions are typically loaded onto computer **601** to cause a series of operational steps to be performed by processor set **610** of computer **601** and thereby effect a computer-implemented method, such that the instructions thus executed will instantiate the methods specified in flowcharts and/or narrative descriptions of computer-implemented methods included in this document (collectively referred to as “the inventive methods”). These computer readable program instructions are stored in various types of computer readable storage media, such as cache **621** and the other storage media discussed below. The program instructions, and associated data, are accessed by processor set **610** to control and direct performance of the inventive methods. In computing environment **600**, at least some of the instructions for performing the inventive methods may be stored in block **650** in persistent storage **613**.

[0045] COMMUNICATION FABRIC **611** is the signal conduction path that allows the various components of computer **601** to communicate with each other. Typically, this fabric is made of switches and electrically conductive paths, such as the switches and electrically conductive paths that make up busses, bridges, physical input/output ports and the like. Other types of signal communication paths may be used, such as fiber optic communication paths and/or wireless communication paths.

[0046] VOLATILE MEMORY **612** is any type of volatile memory now known or to be developed in the future. Examples include dynamic type random access memory (RAM) or static type RAM. Typically, volatile memory **612** is characterized by random access, but this is not required unless affirmatively indicated. In computer **601**, the volatile memory **612** is located in a single package and is internal to computer **601**, but, alternatively or additionally, the volatile memory may be distributed over multiple packages and/or located externally with respect to computer **601**.

[0047] PERSISTENT STORAGE **613** is any form of non-volatile storage for computers that is now known or to be developed in the future. The non-volatility of this storage means that the stored data is maintained regardless of whether power is being supplied to computer **601** and/or directly to persistent storage **613**. Persistent storage **613** may be a read only memory (ROM), but typically at least a portion of the persistent storage allows writing of data, deletion of data and re-writing of data. Some familiar forms of persistent storage include magnetic disks and solid state storage devices. Operating system **622** may take several forms, such as various known proprietary operating systems or open source Portable Operating System Interface-type operating systems that employ a kernel. The code included in block **650** typically includes at least some of the computer code involved in performing the inventive methods.

[0048] PERIPHERAL DEVICE SET **614** includes the set of peripheral devices of computer **601**. Data communication connections between the peripheral devices and the other components of computer **601** may be implemented in various ways, such as Bluetooth connections, Near-Field Communication (NFC) connections, connections made by cables (such as universal serial bus (USB) type cables), insertion-type connections (for example, secure digital (SD) card), connections made through local area communication networks and even connections made through wide area networks such as the internet. In various embodiments, UI device set **623** may include components such as a display screen, speaker, microphone, wearable devices (such as goggles and smart watches), keyboard, mouse, printer, touchpad, game controllers, and haptic devices. Storage **624** is external storage, such as an external hard drive, or insertable storage, such as an SD card. Storage **624** may be persistent and/or volatile. In some embodiments, storage **624** may take the form of a quantum computing storage device for storing data in the form of qubits. In embodiments where computer **601** is required to have a large amount of storage (for example, where computer **601** locally stores and manages a large database) then this storage may be provided by peripheral storage devices designed for storing very large amounts of data, such as a storage area network (SAN) that is shared by multiple, geographically distributed computers. IoT sensor set **625** is made up of sensors that can be used in Internet of Things applications. For example, one sensor may be a thermometer and another sensor may be a motion detector.

[0049] NETWORK MODULE **615** is the collection of computer software, hardware, and firmware that allows computer **601** to communicate with other computers through WAN **602**. Network module **615** may include hardware, such as modems or Wi-Fi signal transceivers, software for packetizing and/or de-packetizing data for communication network transmission, and/or web browser software for communicating data over the internet. In some embodiments, network control functions and network forwarding functions of network module **615** are performed on the same physical hardware device. In other embodiments (for example, embodiments that utilize software-defined networking (SDN)), the control functions and the forwarding functions of network module **615** are performed on physically separate devices, such that the control functions manage several different network hardware devices. Computer readable program instructions for performing the inventive methods can typically be downloaded to computer **601** from an external computer or external storage device through a network adapter card or network interface included in network module **615**.

[0050] WAN **602** is any wide area network (for example, the internet) capable of communicating computer data over non-local distances by any technology for communicating computer data, now known or to be developed in the future. In some embodiments, the WAN **602** may be replaced and/or supplemented by local area networks (LANs) designed to communicate data between devices located in a local area, such as a Wi-Fi network. The WAN and/or LANs typically include computer hardware such as copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and edge servers.

[0051] END USER DEVICE (EUD) **603** is any computer system that is used and controlled by an end user (for example, a customer of an enterprise that operates computer **601**), and may take any of the forms discussed above in connection with computer **601**. EUD **603** typically receives helpful and useful data from the operations of computer **601**. For example, in a hypothetical case where computer **601** is designed to provide a recommendation to an end user, this recommendation would typically be communicated from network module **615** of computer **601** through WAN **602** to EUD **603**. In this way, EUD **603** can display, or otherwise present, the recommendation to an end user. In some embodiments, EUD **603** may be a client device, such as thin client, heavy client, mainframe computer, desktop computer and so on.

[0052] REMOTE SERVER **604** is any computer system that serves at least some data and/or functionality to computer **601**. Remote server **604** may be controlled and used by the same entity that operates computer **601**. Remote server **604** represents the machine(s) that collect and store helpful and useful data for use by other computers, such as computer **601**. For example, in a hypothetical case where computer **601** is designed and programmed to provide a recommendation based on historical data, then this historical data may be provided to computer **601** from remote database **630** of remote server **604**.

[0053] PUBLIC CLOUD **605** is any computer system available for use by multiple entities that provides on-demand availability of computer system resources and/or other computer capabilities, especially data storage (cloud storage) and computing power, without direct active management by the user. Cloud computing typically leverages sharing of resources to achieve coherence and economies of scale. The direct and active management of the computing resources of public cloud **605** is performed by the computer hardware and/or software of cloud orchestration module **641**. The computing resources provided by public cloud **605** are typically implemented by virtual computing environments that run on various computers making up the computers of host physical machine set **642**, which is the universe of physical computers in and/or available to public cloud **605**. The virtual computing environments (VCEs) typically take the form of virtual machines from virtual machine set **643** and/or containers from container set **644**. It is understood that these VCEs may be stored as images and may be transferred among and between the various physical machine hosts, either as images or after instantiation of the VCE. Cloud orchestration module **641** manages the transfer and storage of images, deploys new instantiations of VCEs and manages active instantiations of VCE deployments. Gateway **640** is the collection of computer software, hardware, and firmware that allows public cloud **605** to communicate through WAN **602**.

[0054] Some further explanation of virtualized computing environments (VCEs) will now be provided. VCEs can be stored as “images.” A new active instance of the VCE can be instantiated from the image. Two familiar types of VCEs are virtual machines and containers. A container is a VCE that uses operating-system-level virtualization. This refers to an operating system feature in which the kernel allows the existence of multiple isolated user-space instances, called containers. These isolated user-space instances typically behave as real computers from the point of view of programs running in them. A computer program running on an ordinary operating system can utilize all resources of that computer, such as connected devices, files and folders, network shares, CPU power, and quantifiable hardware capabilities. However, programs running inside a container can only use the contents of the container and devices assigned to the container, a feature which is known as containerization.

[0055] PRIVATE CLOUD **606** is similar to public cloud **605**, except that the computing resources are only available for use by a single enterprise. While private cloud **606** is depicted as being in communication with WAN **602**, in other embodiments a private cloud may be disconnected from the internet entirely and only accessible through a local/private network. A hybrid cloud is a composition of multiple clouds of different types (for example, private, community or public cloud types), often respectively implemented by different vendors. Each of the multiple clouds remains a separate and discrete entity, but the larger hybrid cloud architecture is bound together by standardized or proprietary technology that enables orchestration, management, and/or data/application portability between the multiple constituent clouds. In this embodiment, public cloud **605** and private cloud **606** are both part of a larger hybrid cloud.

[0056] FIG. **7** illustrates a block diagram of components **751-753** included in the pressurized sleeve for linear component installation/uninstallation program **650**, in accordance with an exemplary embodiment of the present inventive concept.

[0057] The pressurized sleeve for linear component installation/uninstallation program **650** can include an obtainment component **751**. The obtainment component **751** can obtain project data related to a project that includes installation and/or uninstallation of at least one linear component **101** (e.g., at least partially flexible) based on at least one of user input and/or predetermined extracted features signifying the project from the obtained project data. The obtainment component **751** can obtain real-time, published, and/or prior project data (e.g., linear component data and/or project environment data, etc.) from various sources, including but not limited to: user input, prior machine learning (e.g., similar/identical projects, project environments **400**, and/or predetermined project elements, etc.), the IoT feed, the at least one 3D depth camera **406**, the 3D models, the controller **304**, the pressurized sleeve **100**, the sensors **103** (e.g., the bend sensors **103***b*, the strain gauges **103***a*, the hydrostatic pressure sensors, the connection orientation fixture **108**, connection orientation fixture lasers **109**, etc.), operation logs, connected computing devices (e.g., the external computing device **405**, a manufacturing extraction system, the controller **304**, other robotic devices involved in the implementation of the project, relevant repositories, databases, inventories, etc.), via the network (e.g., web search for published project materials/manuals/instructions/project designs/recalls/disclaimers/reviews, etc.), and/or project design implementation simulations. The obtainment component **751** can obtain project element (e.g., apparatus for installation and/or uninstallation, linear component **101**, other components, robotic cabling devices (e.g., the pressurized sleeve **100**), sub-components, human operators, etc.) characteristics and/or project environment **400** characteristics by extracting features from the respective obtained project data with relevant machine learning processes. The extracted project element characteristics can include, but is not limited to project element: installation/uninstallation means types (e.g., robotic cabling device (e.g. the pressurized sleeve **100**), the robotic stand **401**, sub-components thereof, and/or human operators, etc.); installation/uninstallation means capacities/limitations (e.g., mobility, accuracy, efficiency, delicacy, precision, forces, reach, etc.); types of linear components **101**, sub-components, and/or other components/project elements (e.g., wires, connectors, cables, ropes, rods, beams, clasps, ties, type of apparatus for installation and/or uninstallation (e.g., mainframe, pulley, gondola, support structure, etc.), robotic cabling tools and/or mobility mechanisms thereof, etc.); properties (e.g., mechanical, chemical, electrical, etc.); project element structures (e.g., flexibility, sub-components, layers, dimensions (e.g., 3D shapes, lengths, widths, circumferences, diameters, bends, curves, angles, shapes, etc.), relationships/connectivity/function, etc.); compositions (e.g., materials, compounds, elements, etc.); project element conditions (e.g., age, use duration, magnitudes of prior use (if any), damage, maintenance, replacement, efficiency, etc.); use thresholds/requisites/ranges/maximums/minimums (e.g., electrical (e.g., voltage, resistance, capacitance, etc.), mechanical (e.g., hydrostatic pressures, tensions, bends (e.g., minimum bend radius), compressions, torsion, and/or shearing), thermal (e.g., internal/external/adjacent/regional/positional temperatures), connection forces (e.g., plug forces), lifespan (e.g., durability, arrangement predisposition of project elements (e.g., outer sub-components, high wear-and-tear project elements, etc.), predetermined ages/use durations/use magnitudes, predetermined effects of deviations, etc.); and/or project element installation/uninstallation connection mechanisms (e.g., routing, plugging/unplugging, affixing/removing, soldering/melting, tethering/untethering, etc.), etc.

[0058] The extracted project environment characteristics can include a project: type, minimum/maximum/predetermined space and/or dimensions thereof; ambient/internal/external temperatures, design (e.g., included project elements, orders, 3D models, project element arrangement/groups, project element occupied/unoccupied space, predetermined buffer spaces, routes, bends, angles, curvatures, axial movements, 3D spatial movements/positions, connection points (e.g., plug sites, destinations, etc.), segmented route dimensions/orientations/3D movements/positions, predetermined means/mechanisms of installation/uninstallation, instructions, predetermined human operator vs. automated steps, predetermined parameters, predetermined installation and/or uninstallation sequences, predetermined potential project element substitutes, predetermined project element priorities, etc.); status (e.g., allocated/mobilized/installed/uninstalled/substituted/replaced project elements, stages/steps of completion and/or relative predetermined durations/times, complications, modifications, thermal/electrical/mechanical/operation values/risks, etc.); and/or project element inventories.

[0059] For example, the obtainment component **751** can identify a new project (i.e., mainframe assembly including cable routing) received from feature extraction and analysis of project data received from a connected external computing device **405**. The obtainment component **751** can extract features such as project elements, a preliminary project design, project element orders, project type, parameters/instructions, flexible linear components **101** (i.e., cable), other components (i.e., other cables, fixtures, fasteners, etc.), cable minimum bend radius, cable dimensions, an apparatus for installation/uninstallation (i.e., a mainframe), etc.) and a predetermined project environment **400**. The obtainment component **751** can further obtain live project data from the project environment **400** via in situ IoT feed and 3D depth cameras **406**, extract features of the apparatus (e.g., the mainframe) and dimensions thereof, the pressurized sleeve **100** (i.e., the robotic cable sleeve **100**), the affirmative presence of the robotic stand **401**, the cabling workspace **404**, and/or dimensions thereof.

[0060] The pressurized routing linear components program **650** can include an analysis component **752**. The analysis component **752** can obtain a project design (e.g., preliminary, modified, and/or optimal) via user input, the extracted features, and/or generation based on an analysis of the extracted project element characteristics and/or the extracted project environment characteristics. The analysis component **752** can interpolate the extracted element characteristics and/or the extracted project environment characteristics and perform calculations/simulations with precise values to identify project designs (e.g., routes, 3D spatial positions/movement/shapes, distances, connection points, angles, curvatures, orientations, segments, bends, etc.), implementations, and/or actual/predicted project risks (e.g., deviations in predetermined use parameters, damage, spatial conflicts, and/or functionality of project elements, resource sufficiency, interference between robotic cabling devices and/or human operators, project apparatus efficiency/lifespan, etc.), etc. The generated project design can be at least partially based on mitigating the calculated risks; predetermined project parameters (e.g., predetermined time, project element priorities, cost, etc.); feasibility; installation and/or uninstallation/connection points; requisite connection forces, and/or optimal project element arrangements/configurations/pairs/groups/movement/positions/orientations/angles/connections from among a plurality of project element arrangements/configurations/pairs/groups/movement/positions/orientations/angles/connections (e.g., primary routes, secondary routes, tertiary routes, etc.) based on analysis on the analyzed extracted project element characteristics and/or the extracted project environment characteristics (e.g., occupied/unoccupied space and/or a calculated buffer margin of predicted inevitable deviation therefor, dimensions, bends, curvatures, potential for damage to functionality, efficiency, and/or lifespan, priority, inventory, installation/uninstallation positions, etc.), and/or installation/uninstallation simulations (e.g., predicted space conflicts, incurred damage, loss of functionality, etc.)). The analysis component **752** can calculate required pouch **102** arrangements/configurations (e.g., groups/pairs/arrangements/configurations/positions/regions, etc.), and requisite hydrostatic pressures in the pressurized sleeve **100** necessary to implement the project design, step thereof, transient/permanent 3D shapes, and/or cable routes. The analysis component **752** can generate an interactive virtual display for a user. The interactive display can include a generated 3D model, simulation, extracted features, spatiotemporal risks, analysis annotations, and/or a timelapse of the project design (e.g., installation of the linear component **101**) progress.

[0061] For example, the analysis component **752** can run a simulation of installation of the cable based on the preliminary project design for the assembly of the mainframe based on the analyzed and extracted features from the obtainment component **751**. The analysis component **752** can identify project risks/thresholds/values (e.g., deviations in predetermined use parameters, damage, spatial conflicts, and/or functionality of project elements, resource sufficiency, interference between robotic cabling devices (e.g., the pressurized sleeve **100**, robotic stand **401**, other cabling devices, etc.) and/or human operators, project apparatus outputs, etc.). The analysis component **752** can modify the preliminary project design in accordance with the predetermined project parameters (e.g., predetermined time, project element priorities, cost, etc.), such as by changing installation orders, avoiding collisions between human operators and/or cabling devices, and optimizing cable routing. The optimized cable routing can be at least partially based on analysis of prior installation footage of similar mainframe assembly projects, which consistently resulted in position/route/movement/risk deviations. The analysis component **752** can calculate optimized cable routing from a plurality of potential cable routings based on calculated buffer space, the plug site locations, the cable dimensions, the deviations, project element crowding, and induced cable shapes that avoid incurring bends with a minimum bend radius and a predicted thermal dissipation from other project elements that exceed a threshold value.

[0062] The pressurized routing linear components program **650** can include an implementation component **753**. The implementation component **753** can implement the project design and/or make dynamic modifications/improvisations thereto based on real-time calculations/predictions/precise values, such as by changing the routes, positions, curvatures, angles, movements, bends, and/or 3D shapes of linear components **101** via calculated hydrostatic pressurization of the pouches **102** of the pressurized sleeve **100**. The implementation component **753** can be connected to the project elements, robotic cabling tools, and/or sub-components (e.g., the pressurized sleeve **100**, the controller **304**, the sensors **103**, the manifolds **201**, the manifold openings **206**, manifold apertures, pouches **102**, pumps **104**, compressors **105**, the robotic stand **401**, etc.), the IoT feed, the 3D depth cameras **406**, other project elements (e.g., the project apparatus/cabling workspace **404**, the linear component **101**, other project elements, sub-components, etc.); and/or an external computing device **405** (e.g., a manufacturing execution system, databases, inventories, repositories, etc.), etc. to implement the optimal project design. The implementation component **753** can make dynamic modifications to the implemented optimal project design based on real-time detection/prediction of inevitable deviations (e.g., spatial position/movement/orientation/angle/shape/route, etc.) in the installed/uninstalled project elements and/or calculated risks. The implementation component **753** can analyze/extract features from obtained project data in real-time, such as from IoT feed, the robotic cabling devices/pressurized sleeve **100** and subcomponents thereof (e.g., the sensors **103**, the pouches **102**, the popped pouches **102**, etc.) and/or the at least one 3D depth camera **406** to calculate/predict use deviations, actual/latent risks, and/or actual/latent damage to project elements (e.g., the linear component, other components, the project apparatus, etc.), efficiency, and/or lifespan that would otherwise be imperceivable to a human operator alone. The implementation component **753** can perform adjustments to the optimal project design and/or the preexisting project design based on the calculated/predicted threshold deviations, actual/latent the risks, and/or the actual/latent damage to the project element (e.g., changing linear component **101** and/or other component routes/shapes/angles/orientations/segments, substituting project element types/units, changing applied installation/uninstallation positions, angles, orientations, applied forces and/or durations, changing mechanisms and/or components for installation/uninstallation, etc.). The implementation component **753** can provide real-time updates/suggestions to the interactive display for the user. In an embodiment that includes human operator connection, the implementation component **753** can calculate and implement predetermined hydrostatic pressures in the pouches **102** that can provide shock absorption at terminal ends and/or avoid excessive bending/physical manipulation.

[0063] For example, the implementation component **753** can implement the modified/optimized project plan generated by the analysis component **752**. The implementation component can mobilize the robotic stand **401** to move into a calculated position/angle/orientation/height upon insertion of the cable **101** into the robotic cable sleeve **100**. The implementation component can implement movements/steps to avoid mutual interference of mobilized human operators and/or cabling devices. The implementation component can initiate installation of the cable **101** by dilating manifold openings **206** and applying calculated hydrostatic pressures to air pouches **102** to induce axial movement of the robotic cable sleeve **100** and the at least partially surrounded cable **101** parallel to a plane of a floor of the project environment **400** via the pump **104**. The 3D depth cameras **406** can confirm achievement of an initial position of the robotic cable sleeve **100** and the at least partially surrounded cable **101**. The implementation component **753** can dilate calculated individual manifold openings **206** and apply calculated hydrostatic pressures via the pump **104** and/or the compressor **105** and respective lines **204**/**205** to achieve the calculated optimal cable routing and installation. The implementation component **753** can identify slightly above threshold strain values via strain gauges **103** of the robotic cable sleeve **100** and can offload requisite hydrostatic pressurization in the affected segment of the cable **101** to a calculated quantity/positions of adjacent air pouches **102** without compromising the general cable routing shape. The implementation component **753** can apply a calculated plug force once the connection orientation fixture laser **109** of the connection orientation fixture **108** confirms spatial positions of terminal ends that are adjacent to the respective plug sites. The implementation component can calculate downstream variations due to the adjustment, such as to a subsequent (lower priority) cable **101** and can adjust routes and configurations accordingly.

[0064] FIG. **8** illustrates a flowchart of a method of linear component installation/uninstallation using pressurized sleeve **800**, in accordance with an exemplary embodiment of the present inventive concept.

[0065] The method includes: [0066] obtaining data related to a project (step **801**); [0067] extracting features from the obtained project data, the extracted features include linear component characteristics and project environment characteristics (step **802**); [0068] obtaining a project design, based on at least one of user input and predetermined extracted features from the extracted features from the obtained project data, the project design includes installation or uninstallation of a linear component in relation to an apparatus (step **803**); [0069] calculating pressurization of a plurality of pouches disposed in a pressurized sleeve for the installation or uninstallation of the linear component in relation to the apparatus based on the project design (step **804**); and [0070] implementing the calculated pressurization of the plurality of pouches disposed in a pressurized sleeve (step **805**).

[0071] Based on the foregoing, a pressurized sleeve for routing linear components, a system implementation, and a method therewith have been disclosed. However, numerous modifications, additions, and substitutions can be made without deviating from the scope of the exemplary embodiments of the present invention. Therefore, the exemplary embodiments of the present invention have been disclosed by way of example and not by limitation.

Claims

1. A pressurized sleeve for routing of linear components, comprising: a linear component at least partially surrounded by the pressurized sleeve; and a plurality of pouches disposed in the pressurized sleeve, wherein at least two pouches of the plurality of pouches have different predetermined hydrostatic pressures from one another which induce movement in at least one dimension of at least a portion of the pressurized sleeve and a corresponding portion of the at least partially surrounded linear component.
2. The pressurized sleeve of claim 1, wherein the linear component is at least partially flexible.
3. The pressurized sleeve of claim 1, wherein the linear component is a cable or a wire.
4. The pressurized sleeve of claim 1, wherein at least one of a pump and a compressor are used to adjust the different predetermined hydrostatic pressures.
5. The pressurized sleeve of claim 1, wherein the different predetermined hydrostatic pressures are static or transient.
6. The pressurized sleeve of claim 1, wherein the induced movement includes at least one bend in the at least the portion of the pressurized sleeve and the corresponding portion of the at least partially surrounded linear component.
7. The pressurized sleeve of claim 6, wherein the at least one bend includes a region of tension and a region of compression.
8. The pressurized sleeve of claim 7, wherein the region of tension and the region of compression are disposed substantially parallel.
9. The pressurized sleeve of claim 8, wherein the region of compression and the region of tension create a resistance to excessive bending of the corresponding portion of the at least partially surrounded linear component.
10. The pressurized sleeve of claim 9, wherein the excessive bending is equal to or greater than a predetermined minimum bend radius of the linear component.
11. The pressurized sleeve of claim 10, wherein at least two pouches of the plurality of pouches are calibrated to pop when the excessive bending occurs.
12. The pressurized sleeve of claim 1, wherein the different predetermined hydrostatic pressures are each pneumatic or hydraulic.
13. The pressurized sleeve of claim 4, wherein the pressurized sleeve is connected to a robotic stand, wherein the robotic stand is connected to the pump, the compressor, at least one 3D depth camera, and a controller, and wherein the controller is further connected to a manufacturing execution system.
14. The pressurized sleeve of claim 13, wherein when the at least partially surrounded flexible linear component is loaded into the pressurized sleeve, the controller uses the pump to inflate at least some of the plurality of pouches to induce axial movement such that the linear component is substantially parallel to a plane of a project environment floor.
15. The pressurized sleeve of claim 14, wherein the at least some of the pouches that are inflated cause the pressurized sleeve to extend in at least one dimension to at least one terminal end of the linear component.
16. The pressurized sleeve of claim 4, wherein an installation or an uninstallation of the linear component in relation to an apparatus is performed by the adjusting of the different predetermined hydrostatic pressures based at least in part on a minimum bend radius, linear component route, connection points, and a project design.
17. The pressurized sleeve of claim 1, further comprising: a connector orientation fixture disposed on at least one terminal end of the pressurized sleeve, wherein the connector orientation fixture includes a connection orientation laser for confirming planned spatial positioning.
18. The pressurized sleeve of claim 1, wherein the induced movement includes axial moving, routing, bending, curving, orienting, and angling of at least a portion of the pressurized sleeve and the corresponding portion of the at least partially surrounded linear component.

19. A pressurized sleeve for routing of linear components, comprising: a robotic cable sleeve disposed at least partially surrounding a flexible linear component; and a plurality of pouches and a plurality of sensors disposed in the robotic cable sleeve, wherein at least two of the plurality of pouches have different predetermined hydrostatic pressures from one another that induce or preserve a predetermined shape of at least a portion of the robotic cable sleeve and a corresponding portion of the at least partially surrounded flexible linear component.

20. A pressurized sleeve for routing of linear components, comprising: a robotic cable sleeve disposed at least partially surrounding an at least partially flexible linear component, wherein the at least partially flexible linear component is a wire or a cable, and wherein the robotic cable sleeve includes a plurality of sensors and a plurality of pouches with adjustable hydrostatic pressures; a robotic stand connected to the robotic cable sleeve; a controller connected to the robotic cable sleeve and the robotic stand for implementing a predetermined project design, wherein the implementing of the project design includes adjusting hydrostatic pressures of at least some of the pouches with adjustable hydrostatic pressures of the plurality of pouches with the adjustable hydrostatic pressures, wherein the implementing of the predetermined project design includes installation or uninstallation of the at least partially flexible linear component in relation to an apparatus by inducing predetermined movements, shapes, and positions of the robotic cable sleeve and a corresponding portion of the at least partially flexible linear component; and at least one a 3D depth camera and an IoT feed connected to the controller for obtaining real-time project design characteristics that include the implementation of the predetermined project design, and wherein the controller modifies the predetermined project design based on an analysis of the obtained real-time project design characteristics that include the implementation of the predetermined project design.